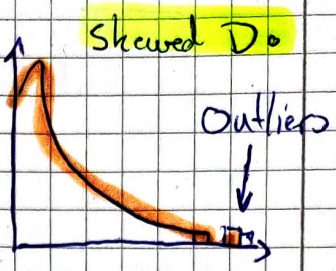
GSB

Data Normalization

→ statistic models often expect data is normally distributed

↳ Data normalization

Skewed Distribution



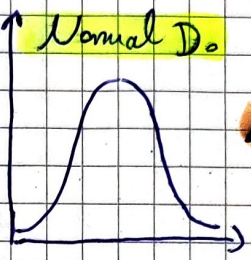
→ exponential distribution

→ lots of extreme values far from mean

→ Hard to do useful statistical tests 😢

↳ Fixing a skewed distribution

log transformed



Z-score

→ Z-score / Standard score:

# of times of standard deviation by which the value of an observation or data point is deviating from the mean value of which is being observed or measured.

Standardizing / normalizing:

$$Z = \frac{X - \mu}{\sigma}$$

GSD

3)

a) 100 results (head / tail) of throwing a coin

Uniform dist

b) 10000 results (1-20) of throwing a 20 sided dice

Uniform dist

c) Number of guest entering between 20-22 over 100 days

Poissian dist

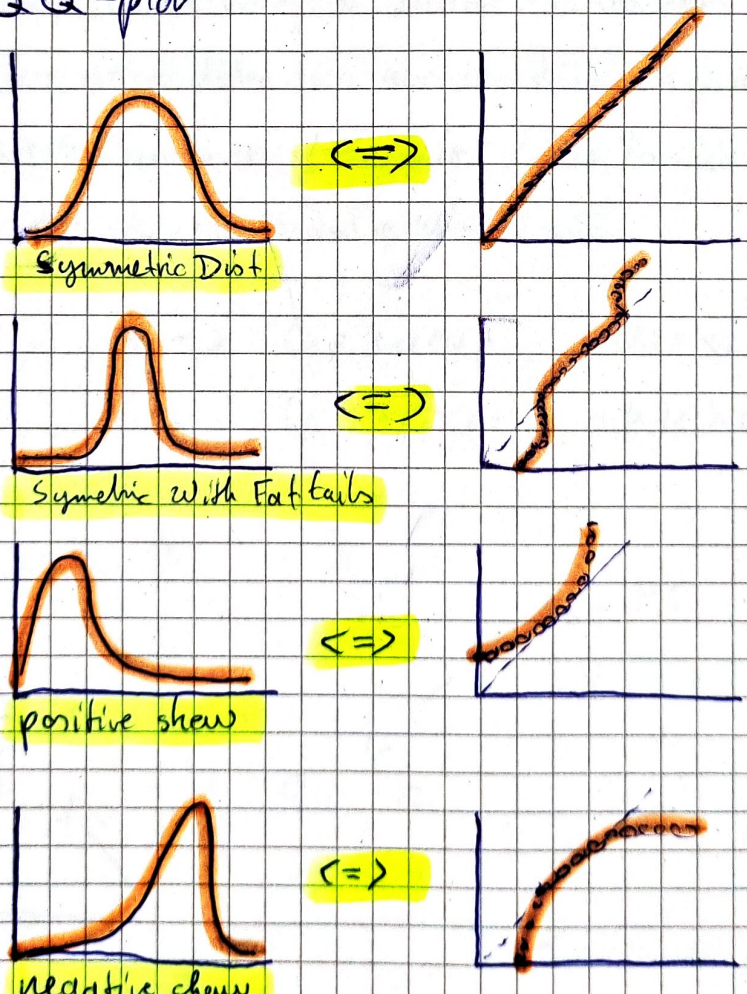
d) IQ human

Normal dist

e) Number of times result head

Binomial dist

4) QQ-plot

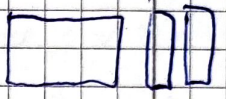


GSB

Principal Component Analysis (PCA)

-> Reduce dimensions/complexity in order to visualize in 2D/3D
-> Visualization of the most dominant signal
-> Detect biological important signal
-> QC: Spot experimental artifacts
-> Identify outliers
-> Use statistical alg to transform a set of possibly correlated variables into a set of uncorrelated linear re combinations of those variables : Principal Component (PC)
-> Capture the most dominant signals and compress it into fewer dimensions. => Capture the most variance of data
-> First Component : most dominating signal
-> PCA is an orthogonal linear transformation
-> transforms data to a new coordinate system, so the greatest variance of the data comes to lie on the first coordinate, then second greatest ...

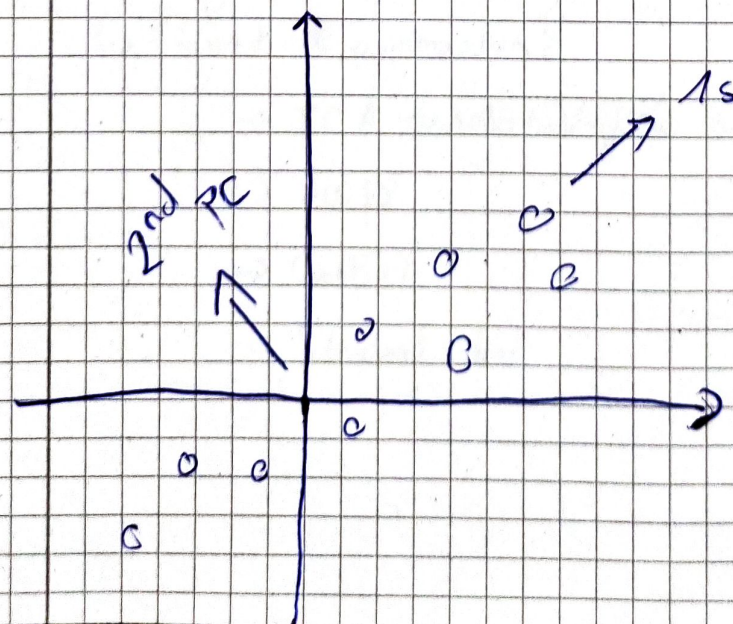
Eigenvektor



original

$x_1, x_2, \ldots, x_n \rightarrow$

$$\left.\begin{aligned} t_1 &= a_{11}x_1 + \ldots + a_{1n}x_n \\ t_n &= a_{n1}x_1 + \ldots + a_{nn}x_n \end{aligned}\right\} \text{uncorrelated}$$



GSB

Batch Effects

-> biased measurements

-> eg: Day to Day Variation

-> Big Challenge in biomedical research -> Missinterpretations

-> Problem:

Batch effects are correlated with an outcome of interest (misses measurement)

-> Bias can be so strong that signal is surpressed

=> Normalization does not remove Batch Effects!

=> Combining different datasets (omics) introduces batch effects!

Avoiding influence of batch effects

-> Beware of how you design your experiment! (large sized cohorts)

-> Solution:

Design to distribute batches and other potential sources of experimental variation across biological group to minimize the probability of confounding between biological and batch effects.

-> Info about changed personal should be recorded

-> Healthy / Unhealthy equal distributed over diff processing times.

- Tackling batch effects bias:

  - Statistical approaches:
    - -> PCA -> substracting the component representing the batch effects
    - -> PVCD
    - -> ComBat
    - -> Standardization

GrSB

DNA Binding Sites

∀ biological functions of DNA are realized by interaction of proteins with specific DNA seq's:

I) Gene Expression by RNA polymerase and associated transcription factors

II) In genome re-arrangements by site-specific recombination and transposition

III) In the restriction and modification of DNA by seq-specific endonucleases / enzymes.

Transcription Factor Binding Sites (TFBS)

-> DNA seq segments that TF bind to => TFBS

-> TF interacts with TFBS with Van der Waals Kräfte.

-> TFs bind to a set of similar DNA seq segments

-> Short: 5-15 bps

-> The seq degeneracy confers different levels of regulation

-> Task: Predict TFBSs

↳ Experimental: -> Traditional: Foot-printing methods
-> High-throughput: SELEX

↳ TFBSs in silico:

↳ aim: to identify more candidate target TFBS

↳ degenerate consensus sequences

↳ Position Weight Matrix (PWM) /
Position specific matrix (PSSM)

GBB

Position Weight Matrices (PWM) Position Specific Scoring Matrices PSSM

PWM is a commonly used representation of motifs in biological seq's.

→ Length of PWM (number of columns): M

↳ Background Model:

$\pi_i \in [0,1]$ with $i \in \{A,T,C,G\}$ and $\sum_i \pi_i = 1$

↳ absolute PWM (count Matrix): ← Build from samples of TF

$f_{ij}^* \in \mathbb{N}$ with $i \in \{A,T,C,G\}$ and $j \in [0, M-1]$

↳ relative PWM (frequency Matrix): ← Papers Zeros

$$f_{ij} = \frac{f_{ij}^*}{\sum_{a \in \{A,T,C,G\}} f_{a,j}^*}$$

Pseudo-counts per column:

$$f_{ij}^P = f_{ij}^* + C\pi_i$$

$$f_{ij} = \frac{f_{ij}^P}{\sum_{a \in \{A,T,C,G\}} f_{a,j}^P}$$

$\pi = BG$	
A	0,23
T	0,25
C	0,25
G	0,27

PWM

FG	1	2	3
A	0	2	2
T	6	0	0
C	1	5	5
G	0	0	0

TCC

PSSM

FG	1	2	3
A	0	2/7	2/7
T	6/7	0	0
C	1/7	5/7	5/7
G	0	0	0

$P(FG \Rightarrow TCC) = \frac{6}{7} \cdot \frac{5}{7} \cdot \frac{5}{7}$ high!

$P(BG \Rightarrow TCC) = 0,25 \cdot 0,25 \cdot 0,25$ low!

GSB

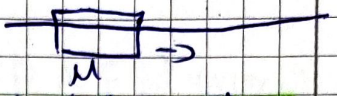
PWM

$$P(AAC, PG) = \frac{f_{A,1}}{\pi_A} + \frac{f_{A,2}}{\pi_A} + \frac{f_{C,3}}{\pi_C}$$

→ Scoring function (log-odd-scores) ↑

$$S_{start\ idx\ / end\ idx} = \sum_{j=start\ idx}^{end\ idx} \ln \frac{f_{nuc(j),j}}{\pi_{nuc(j)}}$$

nuc (j) = nucleotide with index j.



→ Calculate score $S_{0,M-1}$ ∀ position of sliding window

→ Report every match $S_{0,M-1} > th$

↳ What is a good threshold?

Score Distribution

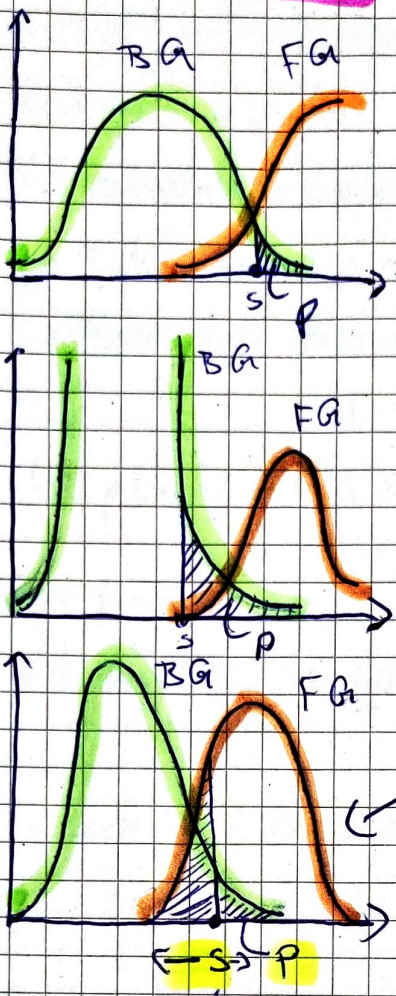
- $\lambda_B(X)$ score distribution of the PWM calculated with random seq's accordg to BG.
- $\lambda_T(X)$ score distribution calculated with random seq's accordg to PWM model. (Signal)
- $P_\lambda(X=s)$ probability of observing score s under distribution λ.

↳ $P_\lambda(X \geq s)$ probability of observing at least score s.

$$P_\lambda(X \geq s) = \sum_{i=s}^{max} P_\lambda(X=i)$$

GxS

Score Distribution



↳ Cutoff ≈ 0,05

Probability by chance

p-Value:

$$p = P_B(X \geq S) = \sum_{i=S}^{max} P_B(X = i)$$

Set $th = S$, with $P_B(X \geq S) = p$

Set equal FP & FN errors:

↳ $P_B(X \geq S) = P_T(X \leq S)$

$p \leq 0{,}05$

↳ H_0 is weak rejected

↳ H_1 stay accepted

GSB

Methods to speed up the general matching

-> Find binding sites by moving a window $O(MN)$

-> Lookahead Algorithm

↳ calculate current score $p[x, S_d]$

↳ calculate worst max possible score σ_d

↳ Intermediate threshold:

$$thd = th - \sigma_d$$

↳ $p[x, S_d] \geq thd \Leftrightarrow S_{0,M-1} \geq th$

-> Even faster:

∀ column has a highest score: $M_j = S_{max(j,j)}$

↳

Expectation of the score by BG:

$$E_j = \sum_a S_{aj} \cdot \pi_a$$

-> Order Matrix by $(M_j - E_j)$ and compute most dangerous first!

GSB-6x1

4)

a) What is a TFBS?

- That location of the DNA where a protein binds to control the transcriptional expression of a gene.

b) What is a PWM?

- A matrix to store the position-specific nucleotide frequencies for a given set of seq's.
- A matrix to store the position-specific amino acid counts for a given set of protein structures.

GSB - Ex 7

cctccaccccc
cctacgccccc
⋮

2) PWMs

a)

PWM:

Position count Matrix

	1	2	3	4	5	6	7	8	9	10
A	3	3	0	2	5	2	0	1	0	0
T	2	0	18	0	0	12	2	0	4	3
C	12	15	0	16	13	1	16	15	16	11
G	0	0	0	0	0	3	0	0	0	4

b) Position relative matrix:

A: 0,2, T: 0,2, C: 0,3, G: 0,3 $\quad C = 4$

	1	2	3	4	5	6	7	8	9	10
A	~~0,12~~	$0,\overline{172}$	...							
T	$0,\overline{172}$	$0,0\overline{36}$	...							
C	$0,\overline{6}$	$0,7\overline{36}$	...							
G	0,1	0,054	...							

↳ Replace zero with

$$f'_{i,j} = \frac{f_{i,j}^{*}}{\sum_k f_{k,j}^{*}}, \quad f_{i,j}^{p} = f_{i,j}^{*} + C \cdot l_i$$

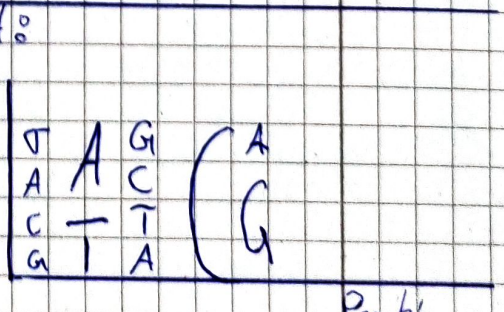
$$f_{i,j} = \frac{f_{i,j}^{p}}{\sum_k f_{k,j}^{p}}, \quad k \in \{A, T, C, G\}$$

~~$f_{0,0} = \frac{3}{1+1+2}$~~

$$f_{0,0}^{p} = 3 + 0,2 \cdot 4 = 3,8$$

$$f_{0,0} = \frac{3,8}{22} = 0,\overline{172}$$

Plot:



Position

3) PCA

- PCA is one of the techniques used for unsupervised learning problems.
- PCA is an orthogonal linear transformation which projects high dimensional data into lower dimensions that capture the most variance.

Multi-Dimensional Scaling (MDS) & PCA?

- For PCA, the input is the scaled (→ Normalized) vectors in n-dimensional space, where the data are projected onto the ~~dd~~ direction with the most variance of the data.
- For MDS, the input is the pairwise distances (between data points) and the output is a two- / three-dimensional projection of the data points, where the distances between the points are preserved.

GSB

Suffix Tree

I) Data structure that represents all the suffixes of a string
II) ∀ suffixes exactly one path in suffix tree
III) Efficient string operations

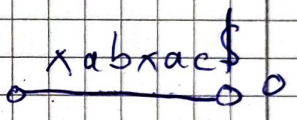
Features: for a string $w[0,..,m-1]$

I) m leaves numbered 0 to $m-1$
II) At least two children for each internal node
III) Each label represents a substring of w
IV) No two edges start with same char

eg:
=> Check if is substring in $O(m)$ and more efficient computations

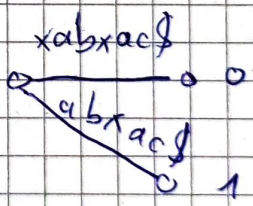
Grow suffix tree

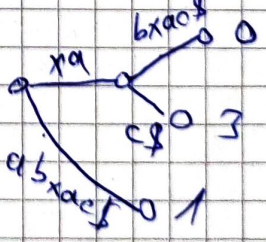
$\rightarrow O(n^2)$



I) Start with the initial string and extend the root to the leaf
II) Generate suffixes $w[1..,n-1]\$$, $w[2..n-1]\$$, ... and push them to the stack
   a) If the matching cannot continue from a node then create new node
   b) If the mismatch occurs in the middle of an edge $e=(u,v)$

abxac$
bxac$
$





GSB

PWM/PSSM using suffix tree

-> Suffix tree improve matching process

I) Allocate Memory

II) Construct Suffix tree from target sequence

III) DFS -> calculate prefix scores $p(x_{sd})$ for edge labels.

~~IV) DFS -> calculate~~

IV) Analyze the scores, check if one of the two holds:

a) Any score at some node in the tree reaches the th, then all of its substrings represented by tree reaches th

b) Check if any of the scores fall below th_d then the whole subtree branch can be ignored

GSB

Suffix tree → Suffix Array

I) suf array : suf array specifies the first ~~char~~ indices of each suffix

II) lcp array : lcp array stores the length of the largest common prefix of two adjacent suffixes according to leaf number.

III) shp array :

1) Sort lexicographically

i	suf[i]	sufix $S_{suf}[i]$
6	0	c d a a a c $
0	1	a a a e c a $

↳ a < c
layer < shorter

II) lcp

i	lcp[i]	suffix
0	0	a a a a c c a c $
1	3	a a a c c a c $
2	2	a a c c a c $

↳ lcp[0] = 0

↳ lcp[i] stores the length of largest common prefix $S_{suf}[i-1]$ & $S_{suf}[i]$

III) shp

shp[i]

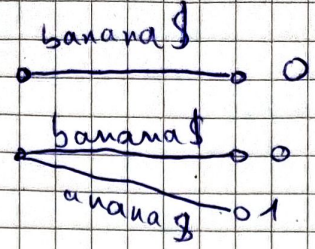
↳ next index j where lcp[j] < lcp[i]

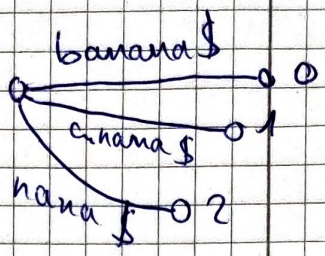
i	lcp[i]	shp[i]	$S_{suf}[i]$
1 2	0	1 2	
	3	2	
	2	?	

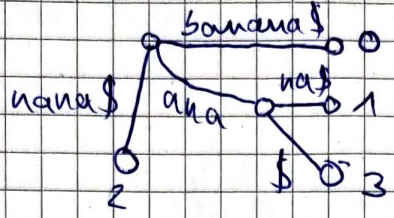
ASB - Ex 8

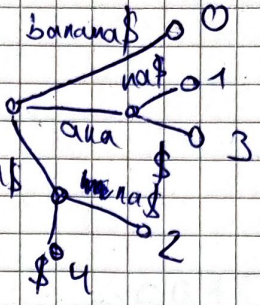
4)

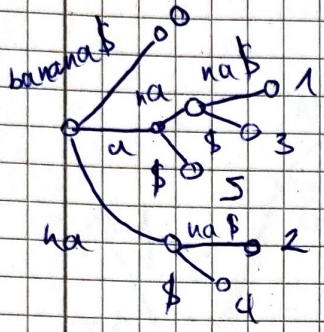
0	banana$
1	anana$
2	nana$
3	ana$
4	na$
5	a$
6	$

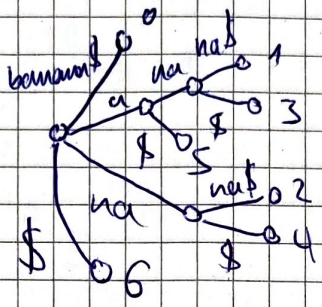




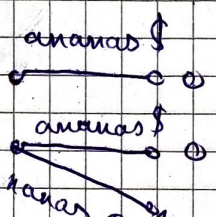


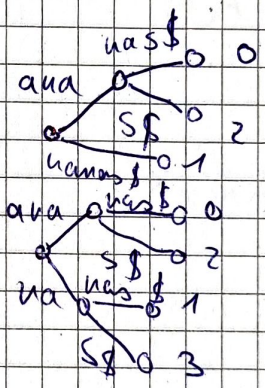


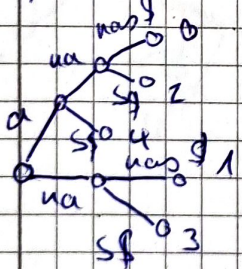


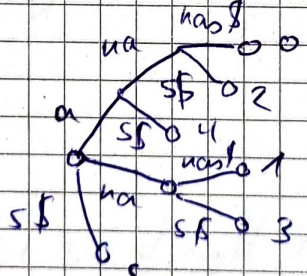


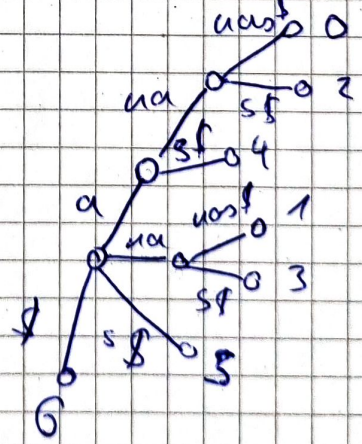
0	ananas$
1	nanas$
2	anas$
3	nas$
4	as$
5	s$
6	$







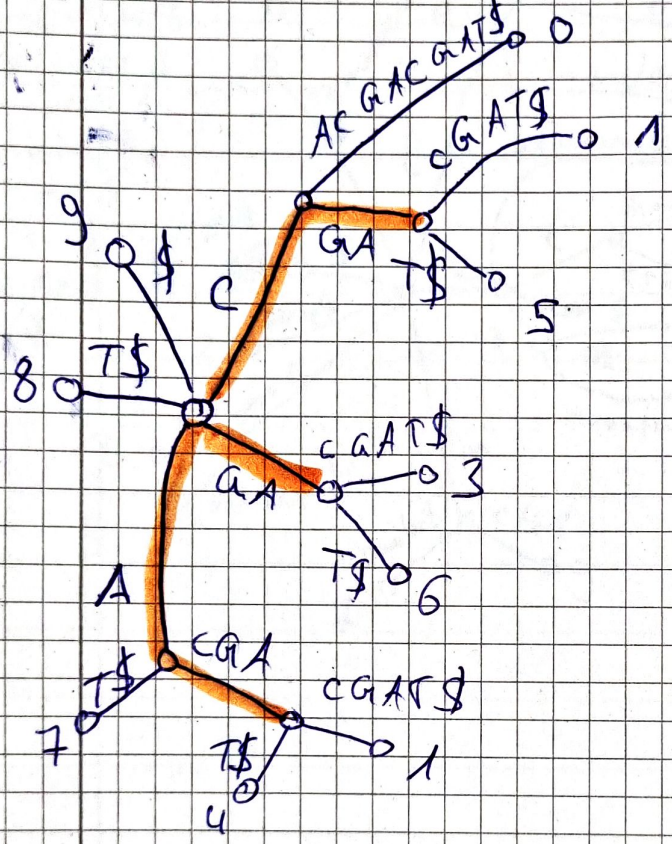




GSS - 68

2) CACGACGAT$

0	CACGACGAT$
1	ACGACGAT$
2	CGACGAT$
3	GACGAT$
4	ACGAT$
5	CGAT$
6	GAT$
7	AT$
8	T$
9	$



CGA { → CGAT, → GAT

GA { → GAT, → T

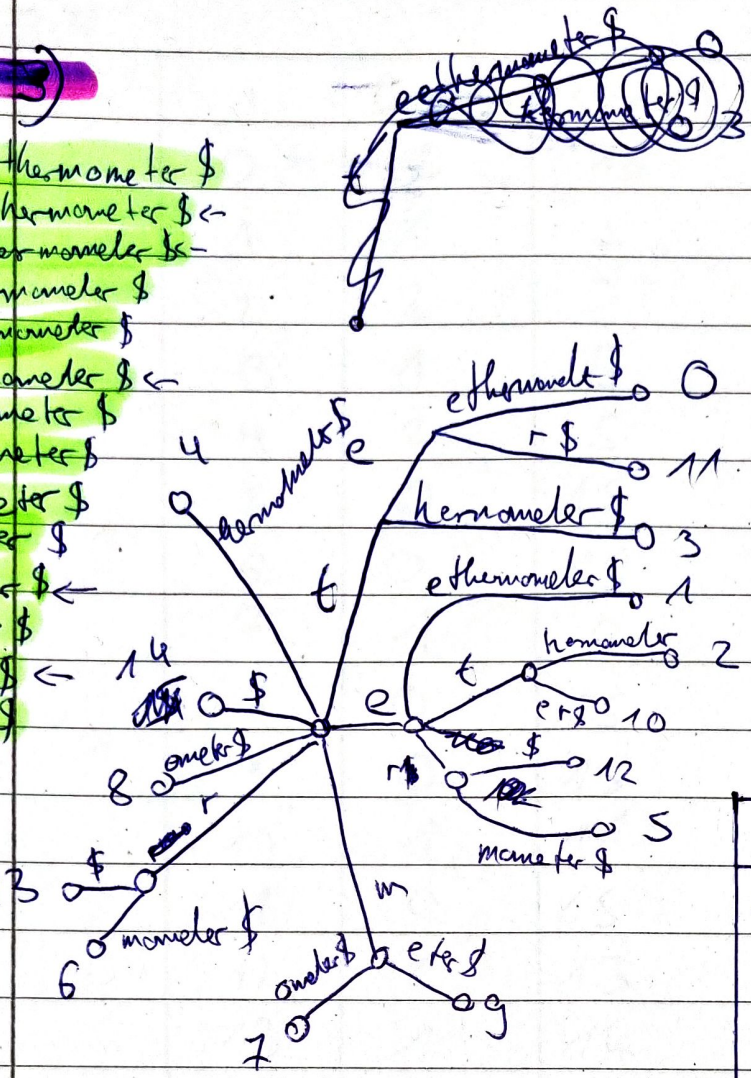
ACGA { → CGAT, → T

= repetitive seq of length ≥ 2

GSB - Ex 8

3)

0 teethermometer $
1 eethermometer $ ←
2 ethermometer $ ←
3 thermometer $
4 hermometer $
5 ermometer $ ←
6 rmometer $
7 mometer $
8 ometer $
9 meter $
10 eter $ ←
11 ter $
12 er $ ←
13 r $
14 $



i	suf[i]	S suf[i]
11	0	teethermometer $
0	1	eethermometer $
4	2	ethermometer $
13	3	thermometer $
5	4	hermometer $
1	5	ermometer $
9	6	rmometer $
7	7	mometer $
8	8	ometer $
6	9	meter $
3	10	eter $
12	11	ter $
2	12	er $
10	13	r $
14	14	$

GSB - Ex 8

3)

i	lcp[i]	skp[i]	~~eethermometer~~ Suff[i]
0	0	15	eethermometer $
1	1	5	ermometer $
2	2	3	er $
3	1	5	eter $
4	2	5	ethermometer $
5	0	15	hermometer $
6	0	15	meter $
7	1	8	mometer $
8	0	15	ometer $
9	0	15	rmometer $
10	1	11	r $
11	0	15	teethermometer $
12	2	13	ter $
13	1	14	thermometer $
14	0	15	$

GSB

Population Genetics

→ study of the distributions and changes of allele frequency in a population.

↳ subject to the four evolutionary processes: natural selection, genetic drift, mutation & gene flow

→ Allele: unterschiedlichen Varianten eines Gens an einer bestimmten Stelle auf einem Chromosom.

→ Allelfrequenz: relative Häufigkeit eines bestimmten Allels in einer Population.

Human Genome

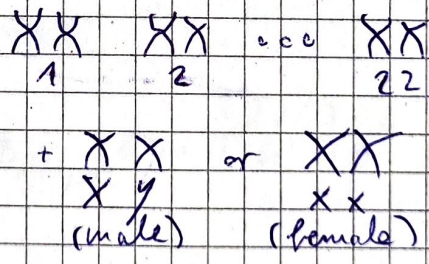
→ 3 billion bases

→ 20 000 - 25 000 genes

→ 2% of the genome encodes genes.

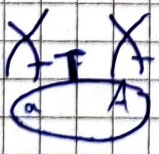
→ 98% no junk: ~~non~~ functionality discovered

→ Chromosome set:

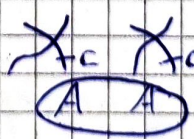


↳ human genome is diploid (2 times identical)

→ Inheritance: only look at 1-22 ~~Chromo~~ autosomes (non-sex)



= heterozygous



= homozygous

genotypes

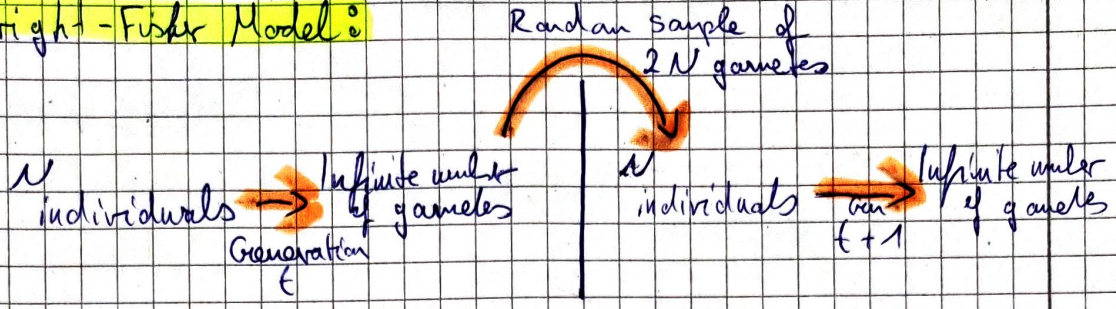
SNP = single-nucleotide polymorphism

GSB

How does Allele frequency change

Gen 0 Gen 1 ... Gen N

Wright-Fisher Model:



→ **Assume:** const size $N < \infty$

& random mating, non-overlapping gens, no migration, no mutation, no selection etc.

N: population size ; n_A, n_a: number of A & a alleles in generation t

At time t:

$$p = \frac{n_A}{2N} \quad \& \quad q = \frac{n_a}{2N} = 1 - p$$

Alleles in generation $t+1$:

$$P\{n_A' = k\} = B_{2N,p}(k) = \binom{2N}{k} p^k (1-p)^{2N-k}$$

⇒ The change of allele frequency from generation to generation in a finite population due to random sampling alone is calle genetic drift.

$0 < p < 1$, the two alleles are segregating

$p = 1$, A allele has been fixed

$p = 0$, A allele has been lost.

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Wright-Fisher-Model

→ Variance of allele frequencies is growing:
↳ Genetic drift causes allele frequencies to drift apart.

→ How quick?
→ Binomial Distributed $B_{n,p}$:

Mean: np

Variance: $np(1-p)$

Standard Deviation: $\sqrt{np(1-p)}$

Allele freq in gen $t+1$

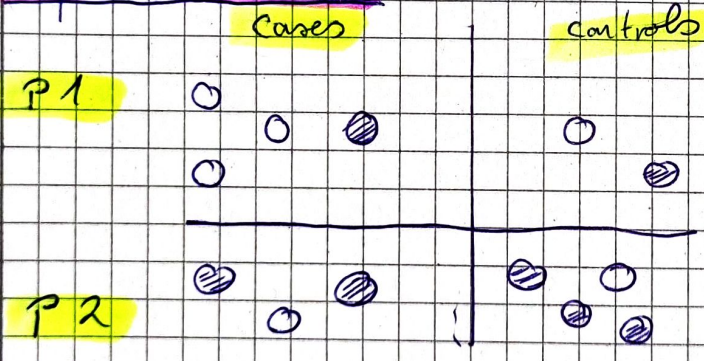
$$\Rightarrow p' = (2Np \pm \sqrt{2Np(1-p)})/2N$$

$$= p \pm \frac{\sqrt{p(1-p)}}{\sqrt{2N}}$$

↳ $N \to \infty$ smaller effect!

→ Evolutionary Pressure: Locus: fester Ort auf Chromosome.
↳ Linkage between two loci: associated with each other?

Population Structure

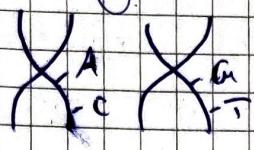


Goal: Avoid false positive hits.

→ Match: BG of cases and control as good as possible
→ Remove SNP positions which are known for ethnic diff's
⇒ Correct for population structure.

GWAS

Phasing



G T

A C

Possible phasings:

G T
A C
G C
A T

A T
G C
A C
G T

Task: • Reconstruct a pair of chromosomes given genotype data for a large # of individuals

• Reconstruct a pair chromosomes given seq data

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Hardy - Weinberg Principle

Distribution of one Genotype

A A
a a
A a

HW - Assumptions:

- → Diploid organisms with sex-reproduction
- → Allele frequency identical for male & female
- → The considered locus has two diff alles
- → Non-overlaping gen's
- → random mating
- → Large (→ ∞) population size
- → No migration, mutation or selection

⇒ Punnet Square:

Male gametes

Female gametes	A (p)	a (q)
A (p)	AA: p^2	Aa: pq
a (q)	aA: pq	aa: q^2

⇒ AA: $P' = p^2$

Aa: $Q' = 2pq$

aa: $R' = q^2$

↓ der Generationswechsel !

Luch a+l $p^2 + pq + q^2$

$P(A) = P(AA) + \frac{1}{2} P(Aa) = \frac{1}{2}(p' + q)$

$P(a) = P(aa) + \frac{1}{2} P(aA) = \frac{1}{2}(q' + q)$

GWAS

Hardy-Weinberg Equilibrium

-> For two alleles A and a and respective allele frequencies p and q, the genotype frequencies P, Q, R for the genotypes AA, Aa, aa are in HWE if $P = p^2$, $Q = 2pq$, $R = q^2$

↳ HWE is stable

↳ In one generation the HWE is reached for two different genotype distributions.

↳ only with HW-Assumptions

Human genetic Variation

eg: Variant A increases the risk for a certain disease.

-> 1000 genome project

↳ Goal: identify genetic causes of disease.

Genome-Wide Association Studies (GWAS)

-> detect disease risk => better monitoring of risk patients

-> Untersuchung der genetischen Variation des Genoms eines Organismus

-> Personalized treatment against a disease

Single Nucleotide Polymorphism (SNP)

additive:
- 0: homozygous major allele (G/G)
- 1: heterozygous (A/G or G/A)
- 2: homozygous minor allele (A/A)

recessive: 0, 0, 2

dominant: 0, 2, 2

GWAS

GWAS: Data

-> ideally full genome

-> SNP-chips: ~600 000 - ~2 500 000 sites

Linkage

-> SNPs in proximity are very often correlated:

eg: $x = A \;\&\; y = G \;||\; x = G \;\&\; y = G$

-> Cause: -> not enough time for evolution

-> Possible lethal effects

=> We need only one SNP per region of SNPs with strong linkage

Single SNP analyses

Phenotype	Genotype 0	1	2
Case	23	41	351
control	104	208	103

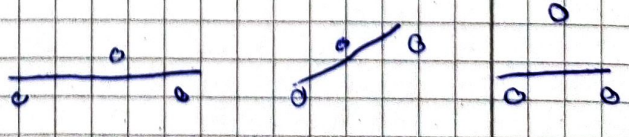
-> Fisher exact test ↳ Tests seeing table bias

Linear Models

-> one allele is considered the risk allele and risks are assumed additive: -> Heterozygous genotype half-way between the two homozygous genotypes

-> Models that test for zero slope for a line that fits the three genotypic risk estimates best: Cochran-Armitage Test

↳ test will not detect the higher disease risk of heterozygotes



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Logestic Regression

- For GWAS

Quantitative Traits (QTs)

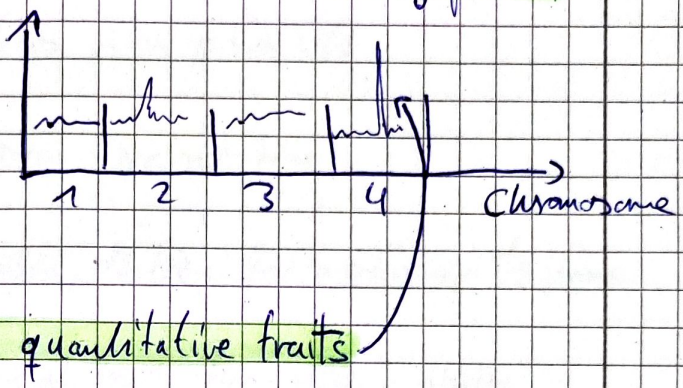
eg: height, longevity, gene expression

→ usually, QTs are determined by an interplay of envir and genetics

→ Many genetic loci contribute to the heritability of QTs

↳ Manhatten P/dt

↳ each dot a SNP



QTLs (QT loci):

↳ The loci that control quantitative traits

↳ eQTL: independent GWAS for expression of each gene

cis-eQTL: position of the eQTL maps near the physical position of the gene.

trans-eQTL: position of the eQTL does not map near the physical position of the gene.

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eQTL concept

→ quantify the amount of RNA in a particular tissue under specific conditions
↳ informative

→ Case:
Different allelic states at a specific locus in the genome alter a measured expression variable in a tissue population under a given set of conditions is an eQTL

↳ pair: genotype - expression association

→ Expression: # RNA molecules, deriving from a position in the genome

→ Polymorphism - the presence of at least two allelic states in a population at a specific locus in the genome where the existence of one of the alleles can be traced to a mutation event.

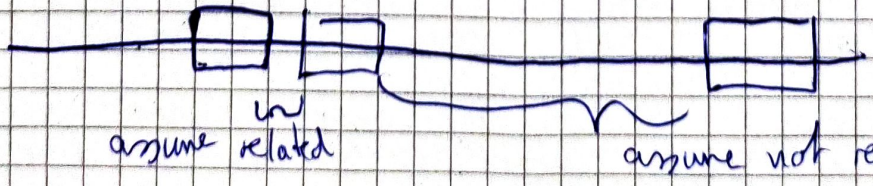
→ Expression Quantitative Trait Locus (eQTL):
a polymorphic locus where an experimental exchange of one allele for another produces a change in expression on average:

$A_1 \rightarrow A_2 \Rightarrow \Delta\bar{y}$ // diff allelic conds

→ The allelic states defined by the original mutation event define the causal polymorphism of the eQTL

→ Test: association between Genotypes!

→ usually near genotypes are related: linkage equilibrium.



GRSB

Multiple Hypothesis Testing

Distribution of p-values:

Uniform under null hypothesis.

→ usually cutoff, $\alpha = 0.05$ → $P(FP) = \alpha$

→ Performing many tests, a fraction of α will have a positive result by chance ⇒ ! control for multiple testing

→ FWER: Family-wise error rate: P for at least one FP

→ FDR: False discovery rate: fraction of FPs among the rejected null hypothesis

Bonferroni Correction

→ Controls FWER

→ N = # of hypothesis to tests ; α = value we want to control the FWER at

→ p_i = p-value for the ith test

⇒ Approach: → p-Value cutoff $\forall$ tests: $\frac{\alpha}{N}$

→ reject H_0 if: $p_i \leq \alpha/N$

→ reduced power when many H's tested

GSD

False Discovery Rate (FDR)

	called not significant	called significant	total
H_0 true	U	V	M_0
H_0 false	T	S	M_1
total	$M-R$	R	M

H_0 true / H_0 false

⇒ We want to find H_0 false

→ False Discovery rate is defined as the ratio of false hypotheses among the ones called significant:

$$FDR = E(V/R)$$

Benjamini-Hochberg Method

$FDR \leq \frac{M_0}{M}\alpha \leq \alpha$ → false discovery rate threshold

Method:

In: α / p-values $p_1, \dots, p_M$

I) Order p-values, s.t. $p_1 < p_2 < \dots < p_M$

II) Compute $L = \max\{j : p_j < \frac{j}{M}\alpha\}$

III) Reject all H_{0j} for which $p_j \leq p_L$

⇒ reject H_0 ! ↳ p significant.

Benjamini-Hochberg Correction

Controls: FDR: fraction of FPs among the rejected H_0's.

I) Sort p-values: $p_1 \leq \dots \leq p_M$

II) For a α, find largest k, $p_k \leq \frac{k}{M}\alpha$

III) Reject H_0's for k tests with the smallest p-values.

↳ large # tests!

↳ Good tradeoff stringency & power

Gr 5 B

1) Sex-specific allele frequencies

two diff genotype distributions for male & female in a population. HW-assumptions

a) until allele frequencies have become equal for male/female.

↳ one generation until they are equal

b) until HWE is reached?

When equal (+1)

↳ 2 Generations until HWE is reached!

⇒ In the absence of selection, mutation, genetic drift, or other forces, allele frequencies p and q are constant between generations!

2) HW for more than one allele A_1, A_2, A_3

a) genotypes

$A_1A_2, A_1A_3, A_2A_3, A_1A_1, A_2A_2, A_3A_3$

↳ #6 types

b)

		A_1 p	A_2 q	A_3 r
A_1	p	p^2	pq	pr
A_2	q	pq	q^2	rq
A_3	r	pr	rq	r^2

⇒ $A_1A_1 = p^2$, $A_1A_2 = 2pq$

$A_2A_2 = q^2$, $A_1A_3 = 2pr$

$A_3A_3 = r^2$, $A_2A_3 = 2qr$

3) Wright-Fisher Model

a) # A_1 & A_2 in 5th gen?

A_1 = 8 , # A_2 = 2

b) # A_1 & A_2 in 6th gen?

A_1 = 8 , # A_2 = 2

c) What is the probability to change from the number of A1 in gen 5 to the number of A1 in gen 6?

$$P_{8,8} \quad \text{Prob} = \binom{10}{8} \left(\frac{8}{10}\right)^8 \left(1-\left(\frac{8}{10}\right)\right)^{10-8}$$

d) - # from 7 to 8?

$$P_{7,7} = \binom{10}{7} \left(\frac{7}{10}\right)^7 \left(1-\left(\frac{8}{10}\right)\right)^{10-7}$$

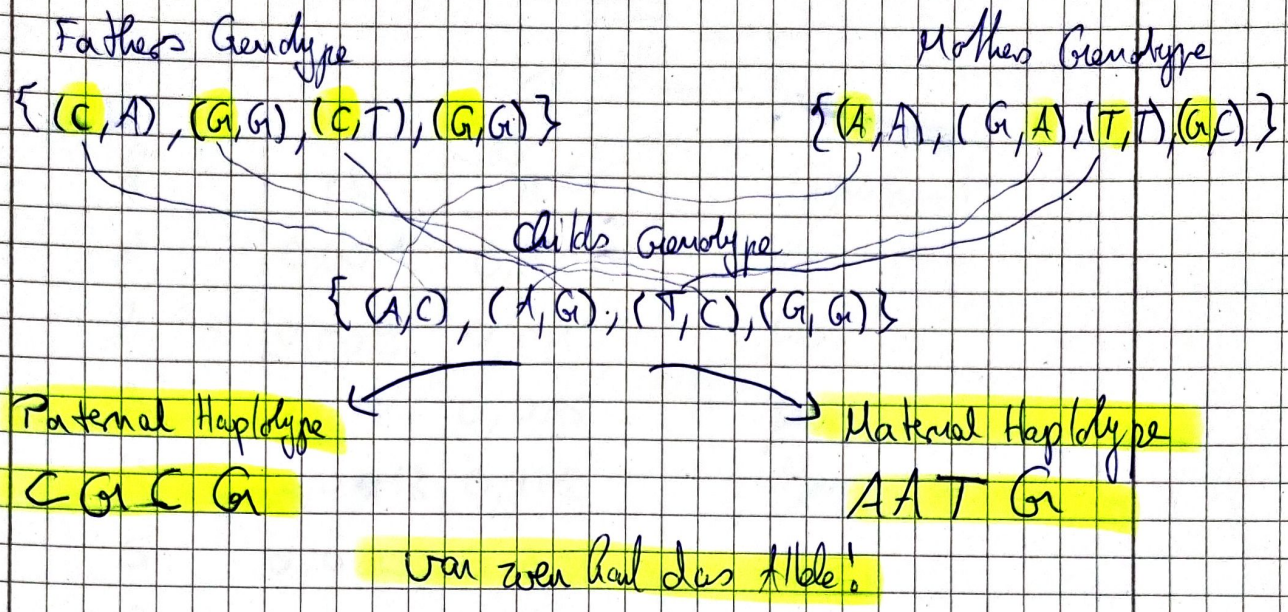
4) Haplotype Phasing

a) $A\{T/G\}\{C/A\}C\,G\{C/A\}$

$\frac{T\,C\,C}{G\,A\,A}$ $\frac{T\,C\,A}{G\,A\,C}$ $\frac{T\,A\,C}{G\,C\,A}$ $\frac{T\,A\,A}{G\,C\,C}$ $\frac{G\,C\,C}{T\,A\,A}$ $\frac{G\,C\,A}{T\,A\,C}$

$\frac{G\,A\,C}{T\,C\,A}$ $\frac{G\,A\,A}{T\,C\,C}$

4) Haplotype Phasing



Von zwei Kopien des Allels!

5) Benjamin Hochberg Correction

$\alpha = 0,05$ → $\frac{a=k}{M=10}\ \alpha = \alpha$

k	p_k	$\frac{k}{M}\alpha$	Significant
1	0,001	0,005	Yes
2	0,009	0,01	Yes
3	0,012	0,015	Yes
4	0,018	0,02	Yes
5	0,025	0,025	Yes
6	0,04	0,03	No
7	0,07	0,035	No
8	0,12	0,04	No
9	0,26	0,045	No
10	0,35	0,05	No

max k with $p_k \leq \frac{a}{M}\alpha$

↳ 5

↳ a-smallest

reject H_0 → significant p